Will be reviewed

8. The Θ expression (tight bound) to describe the number of operations required for the following algorithms is ……..:
   while (n > 1)
   {
    n = n/8; // Use integer division
    } *

- (selected) $\Theta(\log_8 N)$
- $\Theta(N^2 \log_8 N)$
- $\Theta(N \log_8 N)$
- $\Theta(N)$

Will be reviewed

9. Linear Search best case costs O=(N). *

- True
- (selected) False

Will be reviewed

10. The Θ expression (tight bound) to describe the number of operations required for the following algorithms is ……..:
    for (i = 1; i < N; ++i)
    {
    ...statements that require exactly i operations...
    } *

- $\Theta(N \lg N)$
- (selected) $\Theta(N^2)$
- $\Theta(N^3)$

- ○ $\Theta(N)$

Will be reviewed

11. **By applying the construct heap algorithm on the next array,**
15| 16| 15| 7 | 17 | 6
the resulting array will be *

- ◉ $17, 16, 15, 7, 15, 6$
- ○ $17, 15, 16, 7, 15, 6$
- ○ $17, 16, 15, 15, 7, 6$
- ○ $17, 16, 15, 6, 15, 7$

Will be reviewed

12. Question *

$$f(n) = O((f(n))^2).$$

- ◉ $True$
- ○ $Fasle$

Will be reviewed

13. *

$$T(n) = T(n/6) + 2T(n/6) + 3T(n/6) + n^2, \qquad T(n) = O(\ldots\ldots)$$

- ○ $n \lg n$
- ○ $n^2 \lg n$
- ○ $\lg\ n$
- ◉ n^2

Will be reviewed

14. *

$T(n) = 2T(n/2) + 2^{\log_2 n}, \qquad T(n) = O(\ldots\ldots)$

- ○ $n \lg \lg n$
- ◉ $n \lg n$
- ○ n^2
- ○ n

Will be reviewed

15. *

$f(n) = \Theta g(n) \ \ if and only \ \ if \ \ \ g(n) = \Theta f(n).$

- ◉ $True$
- ○ $False$

Will be reviewed

16. *

$T(n) = 2T(\sqrt{n}) + O(\lg n), T(n) = O(\ldots\ldots)$

- ◉ $\lg n \lg \lg n$
- ○ $\lg \lg n$
- ○ n^2
- ○ $n \ \lg n$

Will be reviewed

17. Any decision tree that sorts n elements has height O(n lg n) *

- ○ True
- ◉ False

Will be reviewed

18. The next algorithms use divide and conquer EXCEPT *

- ○ *quick sort*
- ◉ *insertion sort*
- ○ *binary search*
- ○ *heap sort*

Will be reviewed

19. **Give asymptotic upper bound for T(n) in each of the following recurrence.** *

$$2T(n/2) + 4^{\log_4 n}$$

$$n \lg n$$

Will be reviewed

20. *

Quick sort best case running time is $O(n^2)$

- ○ *True*
- ◉ *False*

Will be reviewed

21. Merge sort make sorting in place *

- ○ True

- [x] False

Will be reviewed

22. By applying the insertion sort on the next array,
56 | 82 | 92 | 74 | 76 | 68 | 88 | 68 | 48 | 74 | 18 | 94 ,
how many rounds will be used to sort the array ? *

- [x] 11
- [ ] 10
- [ ] 12
- [ ] n^2

Will be reviewed

23. By applying the quick sort on the next array,
56 | 82 | 92 | 74 | 76 | 68 | 88 | 68 | 48 | 74 | 18 | 94 ,
after the third round of the partitioning the array will be like: *

- [ ] $[18, 48, 56, 92, 74, 76, 68, 88, 68, 74, 82, 94]$
- [x] $[18, 48, 56, 74, 76, 68, 68, 74, 82, 92, 88, 94]$
- [ ] $[18, 48, 56, 92, 74, 76, 68, 88, 68, 74, 94, 82]$
- [ ] $[18, 48, 56, 94, 74, 76, 68, 88, 68, 74, 92, 82]$

Will be reviewed

24. *

$T(n) = 2 * T(n-1) + c$ $T(n) = O(\dots\dots)$

- [ ] $n^2 \lg n$
- [ ] n
- [ ] n^2

- ◉ 2^n

✍ **Will be reviewed**

25. **Give asymptotic upper bound for T(n) in each of the following recurrence.**
T(n) = T(n-1) + n *

n^2

✍ **Will be reviewed**

26. The ……. algorithm uses divide and conquer *

- ◉ *merge sort*
- ○ *insertion sort*
- ○ *selection sort*
- ○ *counting sort*

✍ **Will be reviewed**

27. The Θ expression (tight bound) to describe the number of operations required for the following algorithms is ……..:

```
z = 0 ;
for (w = 0 ; w < K ; w++)
{
      for (x = 0 ; x < N ; x++)
        z = z + x ;
     for (y = 0 ; y < M ; y++)
        z = z + y ;
}
```
*

- ○ $\Theta(\lg N)$
- ◉ $\Theta(N^2)$
- ○ $\Theta(N)$

- ○ $\Theta(N \lg N)$

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